

# Bambinos Marketing & Cohort Intelligence Platform

## PHASE 4 DETAILED IMPLEMENTATION PLAN: DATA GOVERNANCE, INGESTION & PIPELINES

|                                                   |                                                    |
|---------------------------------------------------|----------------------------------------------------|
| Target Standard: v14 Dual-Source Truth Engine     | Target Cadence: Daily 08:00 AM IST (02:30 UTC)     |
| Data Sources: Windsor.ai (Meta Ads) & MariaDB CRM | Orchestration: GCP Cloud Run Jobs + GitHub Actions |

### 1. Executive Overview & Core Objectives

Phase 4 establishes an automated, fault-tolerant data pipeline connecting external Meta Ads delivery facts (via Windsor.ai) with internal CRM bookings, attendance, and invoiced revenue facts in MariaDB. The primary objectives are: (1) eliminate manual report consolidation, (2) guarantee sub-50ms dashboard query latency via pre-aggregated materialized cache tables (cohort\_detail\_cache), and (3) enforce zero-discrepancy checksum reconciliation between ad spend and financial truth.

### 2. Pipeline End-to-End Architecture

| Pipeline Stage     | Component / Module       | Function & Safeguards                                                                                                                                   | Latency |
|--------------------|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| 1. Ingestion       | ingest_windsor.py        | Pulls ad delivery (Spend, Imps, Clicks, LPVs, OS, Placement) from Windsor.ai. Enforces token budgets, 3x backoff, and upserts to raw_meta_ads_delivery. | ~45 sec |
| 2. Normalization   | campaign_tagger.py       | Maps raw campaign strings to canonical Market, Course, and Channel via campaign-tags.csv. Strict exact-matching; unmapped fallback.                     | < 2 sec |
| 3. Materialization | sync_cohort_cache.py     | Executes 28_nightly_cache.sql. Refreshes rolling 35-day cohort window across D0-D30 + Till Date into cohort_detail_cache.                               | ~60 sec |
| 4. Reconciliation  | verify_reconciliation.py | Validates control totals: Spend, Invoiced Revenue, and Lead Counts. Compares raw tables vs cache. Triggers alerts on >0.01% anomaly.                    | < 5 sec |
| 5. Invalidation    | cache_manager.clear()    | Purges backend RAM/file cache, pushes execution telemetry to audit_logger, and exposes fresh state to REST endpoints.                                   | < 1 sec |

### 3. Deep-Dive: Ingestion & Normalization Mechanics

#### Component 1: Windsor.ai Meta Ads Daily Ingestion Engine (ingest\_windsor.py)

- **API Endpoint:** `https://connectors.windsor.ai/meta` with dynamic date-window chunking to prevent API gateway timeouts.
- **Extracted Grain:** `date, account_id, campaign_id, campaign_name, adset_id, ad_id, ad_name, impression_device, publisher_platform`.
- **Metrics Captured:** Spend (Day 0 fixed), Impressions, Link Clicks, Landing Page Views (LPV), and Meta Conversion Actions.
- **Resilience & Cost Controls:** Exponential backoff with jitter (3 retries on 429/500), token usage caps, and idempotent ON DUPLICATE KEY UPDATE upserts.
- **Staging Table DDL (00\_create\_staging\_tables.sql):** High-performance compound primary key on `(date, account_id, ad_id, impression_device, publisher_platform)`.

## 4. Materialization, Quality Control & Automation

### Component 2: MariaDB Materialized Cache Synchronization (sync\_cohort\_cache.py)

- **Stored Procedure Execution:** Calls sp\_refresh\_cohort\_cache\_nightly() compiled from 28\_nightly\_cache.sql.
- **Rolling 35-Day Window:** Recalculates cohorts within the active maturation window so that conversion and attendance events accumulating over D1-D30 are captured accurately.
- **Historical Rebuild Mode:** Supports full multi-year rebuild via --mode full with chunked transactions.

### Component 3: Checksum & Data Integrity Verification Suite (verify\_reconciliation.py)

- **Spend Reconciliation:** Verifies that SUM(spend) in raw\_meta\_ads\_delivery matches SUM(spend) in cohort\_detail\_cache exactly.
- **Revenue Reconciliation:** Cross-checks invoiced new\_revenue in internal billing against cohort total new revenue.
- **Lead & Booking Integrity:** Ensures contact counts and demo counts match internal MariaDB truth.
- **Governance Alert:** Flags any unmapped campaigns exceeding a 2% spend threshold for immediate tagging.

### Component 4: Scheduled Orchestration & Automation

- **Master Runner:** backend/scripts/run\_daily\_pipeline.py chains Ingestion → Cache Sync → Verification → Cache Flush.
- **GCP Cloud Run Job:** deploy\_marketing\_job.sh creates marketing-daily-sync Cloud Run Job in asia-south1 matching bambinos-core conventions.
- **Cloud Scheduler / GitHub Actions:** Scheduled to fire at **08:00 AM IST (02:30 UTC)** daily with manual on-demand trigger support.

## 5. Step-by-Step Implementation & Delivery Roadmap

| Step | Task Description            | Deliverables                                                                 | Validation Method                                                              |
|------|-----------------------------|------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| 4.1  | Windsor.ai Ingestion Engine | ingest_windsor.py<br>00_create_staging_tables.sql                            | Mocked unit tests + dry-run API pull verifying schema mapping & deduplication. |
| 4.2  | Cohort Cache Sync Engine    | sync_cohort_cache.py<br>Stored procedure runner                              | Verify table population and query speed improvement (<50ms).                   |
| 4.3  | Checksum & Quality Gate     | verify_reconciliation.py<br>Control total checkers                           | Automated reconciliation unit tests with artificial discrepancy checks.        |
| 4.4  | Master Runner & Schedulers  | run_daily_pipeline.py<br>daily-marketing-sync.yml<br>deploy_marketing_job.sh | End-to-end execution test in CLI and simulated GitHub Actions run.             |

**Architectural Sign-off:** This plan strictly adheres to the Bambinos v14 analytical contract: Spend is fixed at D0, Leads belong permanently to their capture cohort, and Ratios are never averaged. Upon approval, implementation proceeds systematically through Steps 4.1 to 4.4.